

ARKYA CHATTERJEE

arkyac.github.io

ACADEMIC POSITIONS

C.N.Yang Institute for Theoretical Physics, Stony Brook University *2025 – present*
Research Assistant Professor

EDUCATION

Massachusetts Institute of Technology (MIT) *2019 – 2025*
Ph.D. in Condensed Matter Theory, Department of Physics
Advisor: Xiao-Gang Wen

Indian Institute of Technology Bombay (IIT-B) *2015 – 2019*
B.Tech. (with Honors) in Engineering Physics and Minor in Mathematics

ACADEMIC HONORS

Kendall Teaching Fellow for exemplary performance as a Teaching Assistant, MIT *2025*
C.M.Clay and H.W.Kendall Fellowship, MIT *2019*
K.Seshia Research Excellence Award for best undergraduate thesis, IIT-B Physics *2019*
DAAD WISE fellowship awarded by DFG for a research internship in Germany *2018*
Summer Research Fellowship awarded by the Indian Academies of Science *2017*

TEACHING AND MENTORSHIP

Massachusetts Institute of Technology, Cambridge, MA

- Teaching Assistant for quantum, statistical, and classical mechanics classes at undergraduate & graduate levels; conducted weekly recitations and contributed to setting and grading exams.
- Mentored undergraduate students as part of the Physics Directed Reading Program

Indian Institute of Technology Bombay, Mumbai, India

- Teaching Assistant for first-year undergraduate electromagnetism and calculus classes
- Mentor for the Summer of Science reading projects on advanced physics and math topics

CONFERENCES AND WORKSHOPS

Paths to Quantum Field Theory 2024: invited talk *Jul 2024*
University of Sarajevo

Prospects in Theoretical Physics 2024 summer school: poster & gong show *Jul 2024*
Institute for Advanced Study

Summer school on “Symmetries and Anomalies”: poster & gong show *Jun 2024*
Institut des Hautes Études Scientifiques

Higher Categorical Tools for Quantum Phases of Matter: gong show *Mar 2024*
Perimeter Institute for Theoretical Physics

Simons Ultra-Quantum Matter collaboration meeting: poster
University of Colorado Boulder

May 2023

INVITED TALKS

Physics journal club, Simons Center for Geometry and Physics (SCGP)	<i>Sep 2025</i>
Generalized Symmetries in Quantum Field Theory (GENSYM25) workshop talk Kavli Institute for Theoretical Physics (KITP)	<i>May 2025</i>
String Theory Seminar, Mathematical Institute, University of Oxford	<i>Dec 2024</i>
CQIF seminar, DAMTP, University of Cambridge	<i>Nov 2024</i>
Condensed Matter Seminar (virtual), National University of Singapore	<i>Nov 2024</i>
Quantum Matter in Mathematics & Physics seminar Harvard University Center of Mathematical Sciences and Applications	<i>Jun 2024</i>
Symmetry Seminar (virtual), University of Oxford	<i>May 2024</i>
CMT seminar, Indian Institute of Science	<i>Apr 2024</i>
CMT Kids' seminar, Harvard University	<i>Feb 2024</i>
CMT seminar, Boston University	<i>Nov 2023</i>
Physics colloquium, IIT-B	<i>Jun 2023</i>
CMT seminar (virtual), Pennsylvania State University	<i>Apr 2023</i>
Category and Topological Order seminar (virtual) Beijing Institute of Mathematical Sciences and Applications	<i>Nov 2022</i>

SERVICE AND OUTREACH

Project SHORT , Mentor	<i>2024 – present</i>
Responsible for providing 1-on-1 graduate school application assistance to prospective Physics PhD candidates; aimed at shrinking the socioeconomic gap in graduate school	
Phys. Rev. Lett., Phys. Rev. B, and SciPost Phys. , Referee	<i>2023 – present</i>
Refereed articles on condensed matter and quantum information physics	
MIT Physics Graduate Students Council (PGSC) , Webmaster	<i>2020 – 22</i>
Responsible for maintaining the PGSC website, providing web-related support to the Physics Values Committee	
Harvard Museum of Natural History , Volunteer	<i>2022</i>
Gallery facilitator for Glass Flowers, Maya Civilization, and Sharks exhibits	
IIT-B Maths & Physics Club , Invited Speaker	<i>2020</i>
Gave a (virtual) pedagogical talk to an undergraduate audience on the quantum Hall effect	
IIT-B Physics Academic Mentorship Program , Co-Lead	<i>2018 – 19</i>
Led a team of 12 mentors to provide academic support and facilitate outreach to ~ 200 students	

PUBLICATIONS AND PREPRINTS

The most up-to-date list of my publications can be found on [Google Scholar](#).

- [1] S. D. Pace, M. L. Kim, **A. Chatterjee**, S.-H. Shao, *Parity anomaly from LSM: exact valley symmetries on the lattice*, [arXiv:2505.04684](#) (under review)
- [2] S. D. Pace, **A. Chatterjee**, S.-H. Shao, *Lattice T-duality from non-invertible symmetries in quantum spin chains*, *SciPost Phys.* **18**, 121 (2025) [[arXiv:2412.18606](#)]
- [3] **A. Chatterjee**, S. D. Pace, S.-H. Shao, *Quantized axial charge of staggered fermions and the chiral anomaly*, *Phys. Rev. Lett.* **134**, 021601 (2025) [[arXiv:2409.12220](#)]
- [4] Z. D. Shi, **A. Chatterjee**, *Analytic framework for self-dual criticality in $\mathbb{Z}_k$ gauge theory with matter*, *Phys. Rev. B* **112**, L081111 (2025) [[arXiv:2407.07941](#)]
- [5] **A. Chatterjee**, Ö. M. Aksoy, X.-G. Wen, *Gapped phases and phase Transitions in spin chains with noninvertible symmetries*, *SciPost Phys.* **17**, 115 (2024) [[arXiv:2405.05331](#)]
- [6] I. T. Rosen, S. Muschinske, C. N. Barrett, **A. Chatterjee**, *et al.*, *A synthetic magnetic vector potential in a 2D superconducting qubit array*, *Nat. Phys.* **20**, 1881 (2024) [[arXiv:2405.00873](#)]
- [7] **A. Chatterjee**, W. Ji, X.-G. Wen, *Emergent generalized symmetry and maximal symmetry-topological-order*, *Phys. Rev. B* **112**, 115142 (2025) [[arXiv:2212.14432](#)]
- [8] **A. Chatterjee**, X.-G. Wen, *Holographic theory for continuous phase transitions: emergence and symmetry protection of gaplessness*, *Phys. Rev. B* **108**, 075105 (2023) [[arXiv:2205.06244](#)]
- [9] **A. Chatterjee**, X.-G. Wen, *Symmetry as a shadow of topological order and a derivation of topological holographic principle*, *Phys. Rev. B* **107**, 155136 (2023) [[arXiv:2203.03596](#)]
- [10] M. Chatterjee, **A. Chatterjee**, A. Nandi, A. Sain, *Dynamics and stability of contractile actomyosin ring in the cell*, *Phys. Rev. Lett.* **128**, 068102 (2022) [[arXiv:2007.13441](#)]
- [11] A. Fischer, **A. Chatterjee**, T. Speck, *Aggregation and sedimentation of active Brownian particles at constant affinity*, *J. Chem. Phys.* **150**, 064910 (2019) [[arXiv:1811.05746](#)]